

Fair Division with Indivisible Goods, Chores, and Cake

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(Joint work with Haris Aziz, Simon Mackenzie, and Mashbat Suzuki)

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Motivation

Example (Organizing an event, e.g., conference, workshop, etc.)

- Hotel rooms for students (limited availability) // indivisible goods
- Student Volunteer Program:
 - get a free student registration;
 - take on some light duties. // indivisible chores(?)
- Travel grants // money; divisible good

Research Question

How to **fairly** allocate a mix of divisible and indivisible resources to participants?

Model

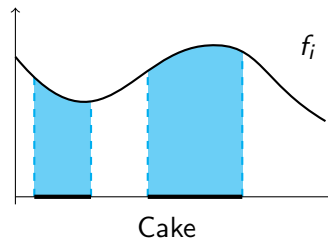
- Agents $N = \{1, 2, \dots, n\}$
- Indivisible items $M = \{1, 2, \dots, m\}$
 - Each agent i has a utility function $u_i: 2^M \rightarrow \mathbb{R}$.
 - We assume **additive utilities**:
 $u_i(M') = \sum_{t \in M'} u_i(t)$ for each bundle $M' \subseteq M$.
 - Allocation: Partition of the goods $(M_1, M_2, \dots, M_n)$.
- A heterogeneous divisible good (or cake) $C = [0, 1]$
 - Agent i has a density function $f_i: [0, 1] \rightarrow \mathbb{R}_{\geq 0}$.
 - Given a piece of cake $S \subseteq [0, 1]$, $u_i(S) = \int_S f_i dx$.
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- Allocation $\mathcal{A} = (A_1, A_2, \dots, A_n)$, where $A_i = M_i \cup C_i$.
- **Utility**: $u_i(A_i) = u_i(M_i) + u_i(C_i)$.

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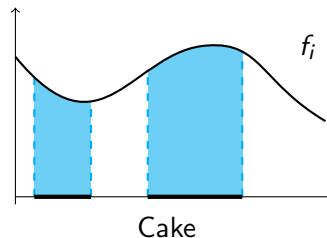
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Envy-freeness for Mixed Goods (EFM)

Definition (EFM [Bei, Li, Liu, Liu, Lu '21])

For any pair of agents i, j ,

- if agent j 's bundle consists of *only* indivisible goods, there exists $g \in A_j$ such that
 $u_i(A_i) \geq u_i(A_j \setminus \{g\});$ // EF1
- otherwise, $u_i(A_i) \geq u_i(A_j).$ // envy-freeness

With only divisible goods: EFM reduces to envy-freeness.

With only indivisible goods: EFM reduces to EF1.

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Theorem (Bei, Li, Liu, Liu, Lu '21)

An EFM allocation always exists for any number of agents.

Proof Idea:

- Start with an EF1 allocation of indivisible goods. // Satisfy EFM.
- Iteratively (and carefully) add some divisible goods to the partial allocation, until all resources are allocated.
- Maintain EFM throughout the process.

EFM (for Goods): Demo

Envy graph: A directed graph of agents with

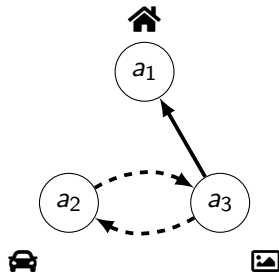
Envy edge: $i \longrightarrow j$ if $u_i(A_i) < u_i(A_j)$;

Equality edge: $i \dashrightarrow j$ if $u_i(A_i) = u_i(A_j)$.

Addable set: A subset of agents $S \subseteq N$ s.t.

- no envy edge in S ;
- no edge from $N \setminus S$ to S .

Envy cycle: A **cycle** in the envy graph with at least one *envy* edge.



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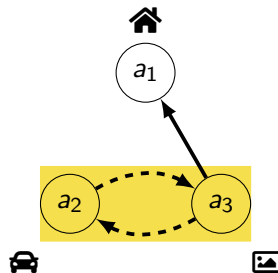
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Identify **disadvantaged** agents.

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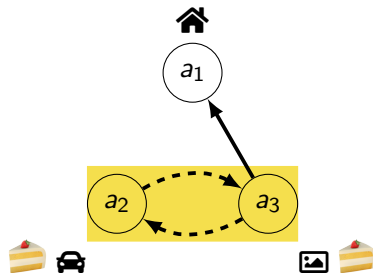
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Add cake to **disadvantaged** agents.

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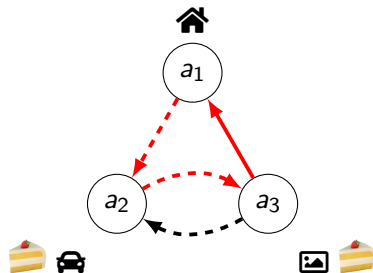
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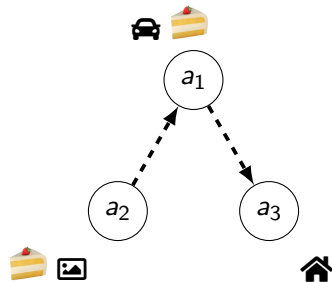
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Eliminate an envy cycle by rotating bundles.

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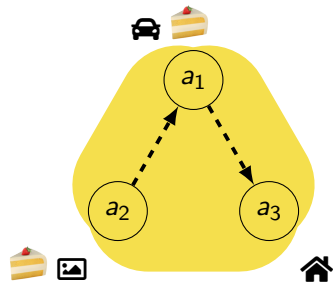
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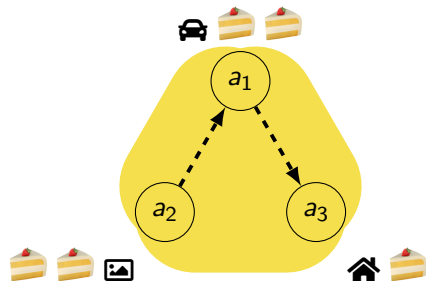
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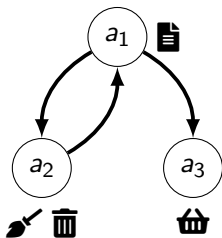


Add cake to **disadvantaged** agents.

Envy-Cycle Elimination Is Problematic When Indivisible Chores Exist

Allocate indivisible chores and a tiny divisible good.

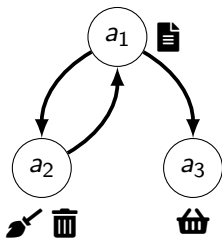
				
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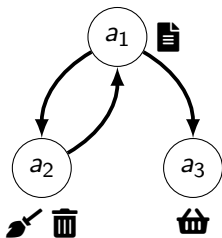


Rotate
→
Bundles

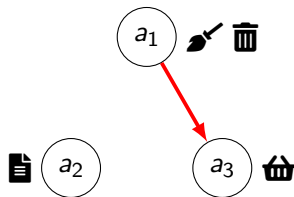
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The envy cannot be eliminated by removing one chore, even if we give a_1 the tiny divisible good!

Existence of EFM When Allocating Mixed Goods and Chores?

Known Results [Bhaskar, Sricharan, Vaish '21]

With indivisible *chores* and divisible goods, EFM allocations exist in two special cases:

- when agents have identical rankings over indivisible chores; or
- when $\# \text{chores} \leq \# \text{agents} + 1$.

Summary of EFM Existence Results

	Cake	Bad Cake (i.e., Divisible Chores)
Indivisible Goods	✓ [Bei et al., '21]	
Indivisible Chores	✓ for very special cases	
Indivisible Goods and Chores	✓ (this talk :)	✓ [Bhaskar et al., '21]

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With indivisible items (consisting of both goods and chores) and cake, an allocation $\{(A_i, C_i)\}_{i \in N}$ is said to be *EFM* if for any pair of agents i, j ,

- either $u_i(A_i) + u_i(C_i) \geq u_i(A_j) + u_i(C_j)$, or // envy-freeness
- both of the following conditions hold:
 - $u_i(C_j) = 0$; and // Agent i values j 's cake at 0
 - $u_i(A_i \setminus \{t\}) \geq u_i(A_j \setminus \{t\})$ for some $t \in A_i \cup A_j$. // EF1
(Remove a chore from i 's bundle, or a good from j 's bundle.)

Note: On the LHS, we did not use " $u_i(A_i \setminus \{t\}) + u_i(C_i)$." Our definition is slightly stronger.

Theorem (Aziz, Lu, Mackenzie, Suzuki '26)

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Inspiration from a Related Line of Research: Fair Division with Subsidy

- Subsidy payments are externally provided.
- Focus on **envy-free** outcomes (allocation of indivisible items plus subsidy payments).
- Care about the worst-case bound on subsidy.

Example (Organizing an event revisited)

Study how much money (given as travel grants) is sufficient to have an envy-free outcome.

Definition (Envy-freeability [Halpern, Shah '19])

An indivisible-items allocation $(A_1, \dots, A_n)$ is said to be **envy-freeable** if there exists a payment vector $(p_1, \dots, p_n)$ such that $\{(A_i, p_i)\}_{i \in N}$ is envy-free, i.e.,

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Cake to Money Reduction

Theorem (Stromquist, Woodall '85)

Let $n \geq 2$ and $(u_i)_{i \in N}$ be utility functions defined over cake $C = [0, 1]$ with $u_i(C) = 1 \forall i \in N$.

Then, **for any** $\alpha \in [0, 1]$, there exists a set $I_\alpha \subseteq C$ such that I_α is the union of at most n intervals and **satisfies** $u_i(I_\alpha) = \alpha$ **for every** $i \in N$.

Theorem (Aziz, Lu, Mackenzie, Suzuki '26)

If there exists an allocation of the indivisible items that is both **EF1** and **envy-freeable**, then an EFM allocation exists with cake.

Proof Idea:



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EF1 Is Compatible with Envy-Freeability in the Goods Setting

Theorem (Brustle, Dippel, Narayan, Suzuki, Vetta '20)

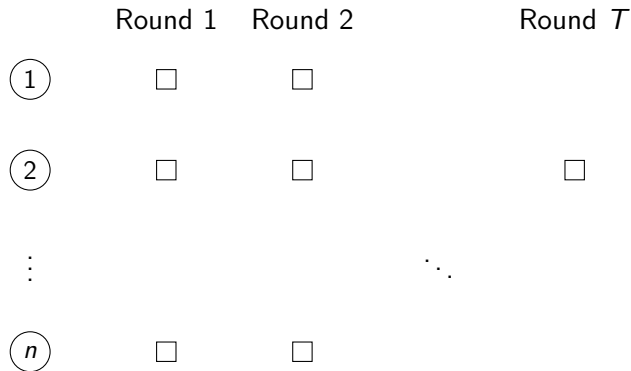
*With indivisible goods, **Iterative Maximum Weight Matching** algorithm outputs an allocation that is both EF1 and envy-freeable.*

Corollary: Together with our previous “Cake to Money Reduction” result, we give a novel algorithm to achieve EFM when allocating a mix of divisible and indivisible goods.

Iterative Maximum Weight Matching (for Goods)

- Bipartite graph between agents and goods.
- Edge (i, g) has weight $u_i(g)$.

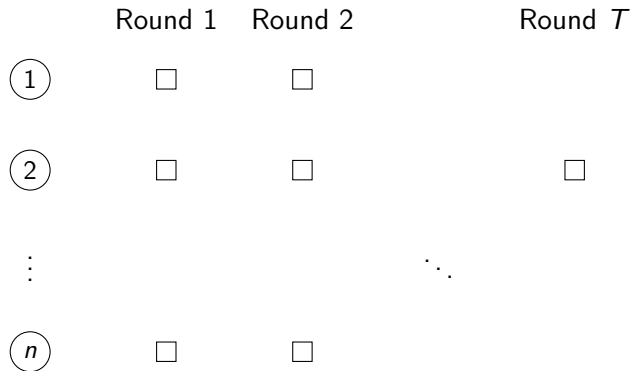
► envy-freeability



Iterative Maximum Weight Matching (for **Subjective Goods**)

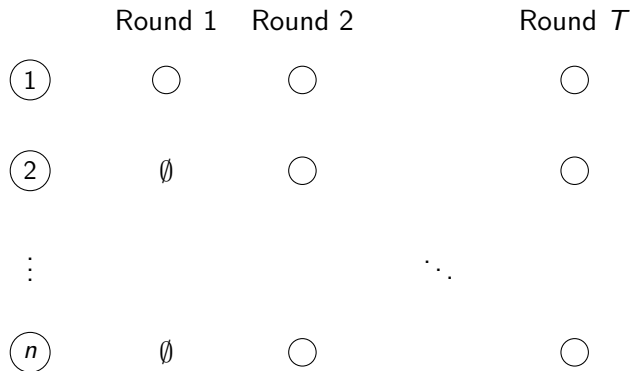
- Bipartite graph between agents and subjective goods.
- Edge (i, g) has weight $u_i(g) \geq 0$.

► envy-freeability



Iterative Maximum Weight *Perfect* Matching (for Objective Chores)

- Add dummy items so that $\# \text{items} = \text{a multiple of } \# \text{agents}$.
- Complete bipartite graph between agents and chores, with edge (i, c) having weight $u_i(c)$.



Bundling

Example (Iterative matching fails without bundling items together)

Two agents have identical utilities over two items: one item has value $+1$; the other has -1 .

Guiding Principles: Bundle items such that running iterative matching algorithms yield EF1 and envy-freeable allocations.

Bundling

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Definition (Meta-good)

We refer to a non-empty subset $M' \subseteq M$ as a **meta-good** if $u_i(M') \geq 0$ for some agent $i \in N$.

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A meta-good M' is said to be **chore-maximal** if for any *objective chore* $c \in M \setminus M'$, we have $u_j(M' \cup \{c\})$ for all $j \in N$.

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Definition (Good-minimality)

A meta-good M' is said to be **good-minimal** if for all $j \in N$ and $g \in \{o \in M' \mid u_j(o) \geq 0\}$, we have $u_j(M' \setminus \{g\}) \leq 0$.

Good-minimality is only needed when objective chores are not enough after initial bundling.

Demo (After Initial Bundling, $\# \text{Chores} \geq n$)

- After initial bundling, we have 2 meta-goods $\diamond$ and $\triangle$, as well as 4 objective chores.
- Agents 1 and 2 value $\diamond$ non-negatively; agent 3 values $\triangle$ non-negatively.

Note: Besides chore-maximality, we further require each agent values at most one meta-good non-negatively during the initial bundling step.

	Round 1	Round 2
①	○	○
②	$\emptyset$	○
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Wrap-up

- Fair allocation of indivisible items (consisting of goods and chores) and a cake.
- Envy-freeness relaxation: EFM // Unify envy-freeness and EF1.
- **Main Result 1:** With indivisible items and cake, an EFM allocation always exists.
- **Main Result 2:** With indivisible items, an EF1 and envy-freeable allocation always exists.

Proof Idea: Carefully bundle items together and run iterative matching algorithms to achieve both EF1 and envy-freeability.

- **Open Question:** Consider utilities more broader than additive utilities.
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Thank You!

Characterization of Envy-Freeability

Theorem (Halpern, Shah '19)

For any allocation $\mathcal{A} = (A_i)_{i \in N}$ of indivisible items, the following statements are equivalent:

- 1 Allocation $\mathcal{A}$ is envy-freeable.
- 2 For every permutation σ over N , we have $\sum_{i=1}^n u_i(A_i) \geq \sum_{i=1}^n u_i(A_{\sigma(i)})$.
- 3 The envy graph $\mathcal{G}_{\mathcal{A}}$ has no positive-weight cycles.

Notation: Envy graph $\mathcal{G}_{\mathcal{A}}$ is a **complete weighted directed graph** over the agents set N :

- each edge (i, j) has weight $u_i(A_j) - u_i(A_i)$ // how much i envies j